WORKED SOLUTION

# STEP II 1996, Question 9

Mechanics • Stability and Centre of Mass

**STEP II 1996, Question 9**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q9

## The key idea

Everything hinges on where the centre of mass $G$ of the toy lies on its axis of symmetry. On the hemisphere, the table always touches the toy directly below the centre $O$ of the hemisphere's flat face, so whether $G$ is below or above $O$ decides between "rights itself" and "topples". On the cone, the toy rests along a whole generator, and it stays put provided the vertical through $G$ meets that line of contact.

## Solution

Model the toy as a rigid uniform solid of density $\rho$, symmetric about its axis. Let $O$ be the centre of the common circular face and measure distances along the axis from $O$, positive towards the vertex $V$ of the cone.

### Centre of mass

The cone has volume $\frac{1}{3}\pi a^2(\lambda a) = \frac{1}{3}\lambda\pi a^3$ and its centre of mass is at $+\frac{1}{4}\lambda a$. The hemisphere has volume $\frac{2}{3}\pi a^3$ and its centre of mass is at $-\frac{3}{8}a$. So $G$ is at

$$h = \frac{\frac{1}{3}\lambda \cdot \frac{1}{4}\lambda a - \frac{2}{3}\cdot\frac{3}{8}a}{\frac{1}{3}\lambda + \frac{2}{3}} = \frac{\frac{1}{12}\lambda^2 a - \frac{1}{4}a}{\frac{1}{3}(\lambda+2)} = \frac{a(\lambda^2-3)}{4(\lambda+2)}.$$

Hence $G$ is below $O$ if $\lambda < \sqrt{3}$, at $O$ if $\lambda = \sqrt{3}$, and above $O$ if $\lambda > \sqrt{3}$.

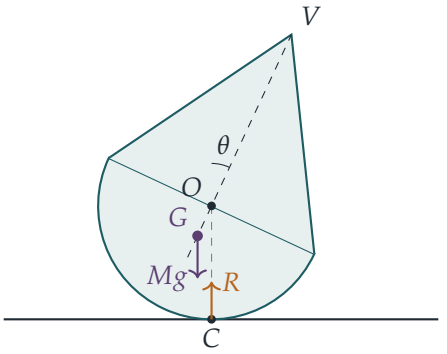
### Resting on the hemisphere

Suppose the axis is tilted at angle $\theta$ to the upward vertical, with $0 \le \theta < \frac{1}{2}\pi$, so that the curved surface of the hemisphere is on the table.

? **Ask yourself** Where exactly does the hemisphere touch the table?

The table is tangent to the sphere of which the hemisphere is part, and a tangent plane

is perpendicular to the radius at the point of contact. So the contact point $C$ is always *directly below* $O$, and $O$ is always at height $a$, whatever $\theta$ is.




(The figure shows $G$ below $O$, the case $\lambda < \sqrt{3}$.)

Take moments about $C$. The normal reaction and any friction act at $C$, so only the weight $Mg$ at $G$ has a moment. The horizontal displacement of $G$ from $C$ (that is, from the vertical through $O$) is $h\sin\theta$, measured towards the side to which the vertex leans.

- If $\lambda < \sqrt{3}$, then $h < 0$. For every $\theta$ in $0 < \theta < \frac{1}{2}\pi$, $G$ lies on the opposite side of the vertical through $C$ from the vertex. So the weight turns the toy back towards $\theta = 0$. Equivalently, $G$ is at height $a + h\cos\theta = a - |h|\cos\theta$, which is least when $\theta = 0$, so the upright position is the position of least potential energy. Either way, wherever it is placed on its hemisphere, the toy moves to the upright position (rocking about it until friction and air resistance settle it there).
- If $\lambda > \sqrt{3}$, then $h > 0$. For any tilt $\theta > 0$, $G$ lies on the same side as the vertex, and the weight turns the toy *further* from upright. The height $a + h\cos\theta$ of $G$ is now greatest at $\theta = 0$, so upright is an unstable equilibrium. The slightest tilt grows until the rim reaches the table and the toy falls onto its conical surface: it overbalances.

(At $\lambda = \sqrt{3}$, $G = O$ and the toy is in neutral equilibrium at every tilt.) This is the required result.

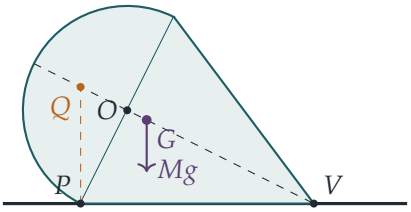
#### Resting on the cone

Now the toy lies with a generator $VP$ of the cone on the table, where $P$ is on the rim. Work in the plane through the axis and $VP$. Take axes in this plane with $O$ at the origin, $V = (0, \lambda a)$ and $P = (a, 0)$, so the hemisphere is on the side $y < 0$. The tangent plane to the cone along $VP$ is perpendicular to this plane, and its trace is the line

$$\frac{x}{a} + \frac{y}{\lambda a} = 1.$$

**Ask yourself** Could the hemisphere poke through the table instead, so that the cone can't lie flat on a generator?

No. Every point of the toy projects into this plane at a point of the cone's triangular section or of the half-disc $x^2 + y^2 \leq a^2$, $y \leq 0$. On the triangle, $\frac{x}{a} + \frac{y}{\lambda a} \leq 1$. On the half-disc, $x \leq a$ and $y \leq 0$, so $\frac{x}{a} + \frac{y}{\lambda a} \leq \frac{x}{a} \leq 1$, with equality only at $P$. So the whole toy lies on one side of the plane, touching it along the segment $VP$ and nowhere else.




(The figure shows $\lambda = 2$. The perpendicular to $VP$ at $P$ is dashed.)

The reactions all act along $VP$ and are perpendicular to the table. So the toy is in equilibrium on the table, and can't tip about $P$ or $V$, exactly when the vertical through $G$ meets the segment $VP$, i.e. when the foot of the perpendicular from $G$ to $VP$ lies between $V$ and $P$.

The perpendicular to $VP$ at $V$ meets the axis only at $V$. Let the perpendicular to $VP$ at $P$ meet the axis at $Q$. In the triangle $VPQ$, the angle at $P$ is a right angle and $PO$ is the altitude to the hypotenuse $VQ$, so $OP^2 = OV \cdot OQ$, giving

$$OQ = \frac{a^2}{\lambda a} = \frac{a}{\lambda}.$$

So $Q$ is a distance $a/\lambda$ below $O$. A point of the axis projects onto the segment $VP$ precisely when it lies between $Q$ and $V$. So we need

$$-\frac{a}{\lambda} < h < \lambda a.$$

The right-hand inequality holds because $G$ is inside the toy, below $V$. For the left-hand one, multiplying by $4\lambda(\lambda+2)/a > 0$,

$$\frac{a(\lambda^2-3)}{4(\lambda+2)} > -\frac{a}{\lambda} \iff \lambda^3 - 3\lambda + 4\lambda + 8 > 0 \iff \lambda^3 + \lambda + 8 > 0,$$

which is true for every $\lambda > 0$. So the vertical through $G$ always meets the line of contact, and the toy stays where it is. By the rotational symmetry, it doesn't matter which generator is in contact either. So whatever the value of $\lambda$, the toy remains on its conical surface, as required.

## Checks

- At $\lambda = \sqrt{3}$ the formula gives $h = 0$: the cone's moment $\frac{1}{12}\lambda^2 a$ exactly balances the hemisphere's $\frac{1}{4}a$. Slicing the toy into thin discs and summing numerically reproduces $h$ to $10^{-5}a$ for several $\lambda$.
- With $\lambda = 1$: $h = -\frac{1}{6}a$ and $OQ = a$, and the foot of the perpendicular from $G$ lies 0.583 of the way from $V$ to $P$. With $\lambda = 10$ it lies 0.790 of the way along. Testing 20,000 random values of $\lambda$ in $(0, 50)$, it always fell strictly inside $VP$.
- Limiting cases: as $\lambda \to 0$ the toy is almost a hemisphere ($h \to -\frac{3}{8}a$), which famously rights itself. As $\lambda \to \infty$ it is a long cone, and $G$ moves up the axis with the cone's own centre of mass.

## Common slips

- Taking moments about the lowest point of the *upright* toy rather than the actual contact point, which moves as the toy rolls but is always directly below $O$.
- In the last part, assuming the hemisphere touches the table too. It doesn't, since the half-disc lies on the side $y \leq 0$.

## Beyond STEP

The rocking toy is a first look at stability of equilibrium. In general, a body rolling on a plane is stable when its centre of mass lies below the centre of curvature of its surface at the contact point (here, $O$). The same idea governs the metacentre of a floating ship. In a first-year course this becomes the study of minima of potential energy and small oscillations about them. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers centres of mass and rigid-body equilibrium in this spirit.

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